

Assignment -4

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1. create a simple car rental program using C#. Your program should model a car rental agency that has a collection of cars, each with a make, model, rental price, and availability status.

```
using System;
using System.Collections.Generic;

namespace Assignment4
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Assignment 4");
        }
    }
}

namespace Assignment4
{
    internal class Car
    {
        public string Make;
        public string Model;
        public decimal RentalPricePerDay;
        public bool IsAvailable;

        public Car(string make, string model, decimal rentalPricePerDay)
        {
            Make = make;
            Model = model;
            RentalPricePerDay = rentalPricePerDay;
            IsAvailable = true; // every car starts out available
        }

        public void DisplayInfo()
        {
            string status = IsAvailable ? "Available" : "Rented out";
            Console.WriteLine($"{Make} {Model} - ${RentalPricePerDay}/day - {status}");
        }
    }
}

using System.Collections.Generic;

namespace Assignment4
{
    internal class RentalAgency
    {
        private List<Car> cars = new List<Car>();

        public void AddCar(Car car)
        {
            cars.Add(car);
        }
    }
}
```

```

public void DisplayAllCars()
{
    foreach (Car car in cars)
    {
        car.DisplayInfo();
    }
}

public bool RentCar(string make, string model)
{
    foreach (Car car in cars)
    {
        if (car.Make == make && car.Model == model)
        {
            if (car.IsAvailable)
            {
                car.IsAvailable = false;
                Console.WriteLine($"You rented the {make} {model}.");
                return true;
            }
            else
            {
                Console.WriteLine($"Sorry, the {make} {model} is already rented out.");
                return false;
            }
        }
    }

    Console.WriteLine($"No car found matching {make} {model}.");
    return false;
}

public void ReturnCar(string make, string model)
{
    foreach (Car car in cars)
    {
        if (car.Make == make && car.Model == model)
        {
            car.IsAvailable = true;
            Console.WriteLine($" {make} {model} has been returned.");
            return;
        }
    }
}

static void Main(string[] args)
{
    RentalAgency agency = new RentalAgency();

    agency.AddCar(new Car("Toyota", "Corolla", 45.00m));
    agency.AddCar(new Car("Honda", "Civic", 50.00m));
    agency.AddCar(new Car("Ford", "Mustang", 90.00m));

    Console.WriteLine("---- All Cars ----");
    agency.DisplayAllCars();
}

```

```

Console.WriteLine();
agency.RentCar("Honda", "Civic");
agency.RentCar("Honda", "Civic"); // try again — should say already rented
agency.RentCar("Tesla", "Model 3"); // doesn't exist

Console.WriteLine();
Console.WriteLine("---- After Renting ----");
agency.DisplayAllCars();

Console.WriteLine();
agency.ReturnCar("Honda", "Civic");

Console.WriteLine();
Console.WriteLine("---- After Returning ----");
agency.DisplayAllCars();
}

```

```

---- All Cars ----
Toyota Corolla - $45.00/day - Available
Honda Civic - $50.00/day - Available
Ford Mustang - $90.00/day - Available

You rented the Honda Civic.
Sorry, the Honda Civic is already rented out.
No car found matching Tesla Model 3.

---- After Renting ----
Toyota Corolla - $45.00/day - Available
Honda Civic - $50.00/day - Rented out
Ford Mustang - $90.00/day - Available

Honda Civic has been returned.

---- After Returning ----
Toyota Corolla - $45.00/day - Available
Honda Civic - $50.00/day - Available
Ford Mustang - $90.00/day - Available

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```

2. Create a C# program that contains a method that finds large numbers of a group of collections respectively. using System.Collections.Generic;

```

namespace Assignment4
{
    internal class NumberFinder
    {
        public int FindLargest(List<int> numbers)
        {
            if (numbers == null || numbers.Count == 0)
            {
                Console.WriteLine("List is empty or null.");
                return -1;
            }

            int largest = numbers[0]; // assume first is largest to start

            foreach (int number in numbers)
            {
                if (number > largest)
                {
                    largest = number;
                }
            }
        }
    }
}

```

```

    }
}

return largest;
}
}
}
Console.WriteLine();
Console.WriteLine("---- Largest Numbers ----");

NumberFinder finder = new NumberFinder();

List<int> group1 = new List<int> { 34, 78, 12, 99, 45 };
List<int> group2 = new List<int> { -5, -3, -10, -1, -7 };
List<int> group3 = new List<int> { 100, 200, 150, 175 };
List<int> group4 = new List<int>(); // empty list to test the guard clause

Console.WriteLine("Group 1 largest: " + finder.FindLargest(group1));
Console.WriteLine("Group 2 largest: " + finder.FindLargest(group2));
Console.WriteLine("Group 3 largest: " + finder.FindLargest(group3));
Console.WriteLine("Group 4 (empty): ");
finder.FindLargest(group4);

```

```

---- Largest Numbers ----
Group 1 largest: 99
Group 2 largest: -1
Group 3 largest: 200
Group 4 (empty):
List is empty or null.

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Press any key to close this window . . .

```

3. Create a C# console application that demonstrates the usage of interfaces and collections.

```

namespace Assignment4
{
    internal interface ICollegeMember
    {
        string Name { get; set; }
        string Id { get; set; }
        void DisplayInfo();
        string GetRole();
    }
}

namespace Assignment4
{
    internal class Student : ICollegeMember
    {
        public string Name { get; set; }
        public string Id { get; set; }
        public string Major;
        public double GPA;

        public Student(string name, string id, string major, double gpa)
        {
            Name = name;
            Id = id;
            Major = major;

```

```

        GPA = gpa;
    }

    public void DisplayInfo()
    {
        Console.WriteLine($"[Student] {Name} (ID: {Id}) | Major: {Major} | GPA: {GPA}");
    }

    public string GetRole()
    {
        return "Student";
    }
}

namespace Assignment4
{
    internal class Teacher : ICollegeMember
    {
        public string Name { get; set; }
        public string Id { get; set; }
        public string Department;
        public int YearsOfExperience;

        public Teacher(string name, string id, string department, int yearsOfExperience)
        {
            Name = name;
            Id = id;
            Department = department;
            YearsOfExperience = yearsOfExperience;
        }

        public void DisplayInfo()
        {
            Console.WriteLine($"[Teacher] {Name} (ID: {Id}) | Dept: {Department} | Experience: {YearsOfExperience} years");
        }

        public string GetRole()
        {
            return "Teacher";
        }
    }
}

using System.Collections.Generic;

namespace Assignment4
{
    internal class CollegeDirectory
    {
        private List<ICollegeMember> members = new List<ICollegeMember>();

        public void AddMember(ICollegeMember member)
        {
            members.Add(member);
            Console.WriteLine($"[{member.GetRole()} '{member.Name}' added to directory.");
        }
    }
}

```

```

    }

    public void DisplayAll()
    {
        Console.WriteLine();
        Console.WriteLine("---- All College Members ----");
        foreach (ICollegeMember member in members)
        {
            member.DisplayInfo();
        }
    }

    public void DisplayByRole(string role)
    {
        Console.WriteLine();
        Console.WriteLine($"---- {role}s only ----");
        foreach (ICollegeMember member in members)
        {
            if (member.GetRole() == role)
            {
                member.DisplayInfo();
            }
        }
    }

    public ICollegeMember FindById(string id)
    {
        foreach (ICollegeMember member in members)
        {
            if (member.Id == id)
            {
                return member;
            }
        }
        return null; // not found
    }
}

Console.WriteLine();
Console.WriteLine("---- College Directory ----");

CollegeDirectory directory = new CollegeDirectory();

directory.AddMember(new Student("Ravi Kumar", "S001", "Computer Science", 8.5));
directory.AddMember(new Student("Priya Singh", "S002", "Electronics", 7.9));
directory.AddMember(new Teacher("Dr. Mehta", "T001", "Mathematics", 15));
directory.AddMember(new Teacher("Prof. Sharma", "T002", "Physics", 10));

directory.DisplayAll();

directory.DisplayByRole("Student");
directory.DisplayByRole("Teacher");

Console.WriteLine();
Console.WriteLine("---- Search by ID ----");

```

```

ICollegeMember found = directory.FindById("T001");
if (found != null)
{
    Console.WriteLine("Found: ");
    found.DisplayInfo();
}
else
{
    Console.WriteLine("Member not found.");
}

```

```

ICollegeMember notFound = directory.FindById("X999");
if (notFound != null)
{
    notFound.DisplayInfo();
}
else
{
    Console.WriteLine("Member with ID X999 not found.");
}

```

```

---- College Directory ----
Student 'Ravi Kumar' added to directory.
Student 'Priya Singh' added to directory.
Teacher 'Dr. Mehta' added to directory.
Teacher 'Prof. Sharma' added to directory.

---- All College Members ----
[Student] Ravi Kumar (ID: S001) | Major: Computer Science | GPA: 8.5
[Student] Priya Singh (ID: S002) | Major: Electronics | GPA: 7.9
[Teacher] Dr. Mehta (ID: T001) | Dept: Mathematics | Experience: 15 years
[Teacher] Prof. Sharma (ID: T002) | Dept: Physics | Experience: 10 years

---- Students only ----
[Student] Ravi Kumar (ID: S001) | Major: Computer Science | GPA: 8.5
[Student] Priya Singh (ID: S002) | Major: Electronics | GPA: 7.9

---- Teachers only ----
[Teacher] Dr. Mehta (ID: T001) | Dept: Mathematics | Experience: 15 years
[Teacher] Prof. Sharma (ID: T002) | Dept: Physics | Experience: 10 years

---- Search by ID ----
Found:
[Teacher] Dr. Mehta (ID: T001) | Dept: Mathematics | Experience: 15 years
Member with ID X999 not found.

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Press any key to close this window . . .

```